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Subject: Programming with Java	Sem: III AY: 2026-27
Assignment No: 10	
TITLE: Write a Java program to demonstrate the use of abstract classes and abstract methods	

1.Create an abstract Payment class and implement Credit Card and UPI payment classes.

```
J PaymentDemo.java X
J PaymentDemo.java > PaymentDemo
1  abstract class Payment {
2
3      abstract void makePayment(double amount);
4  }
5
6  class CreditCardPayment extends Payment {
7
8      void makePayment(double amount) {
9          System.out.println("Paid Rs." + amount + " using Credit Card");
10     }
11 }
12
13 class UPIPayment extends Payment {
14
15     void makePayment(double amount) {
16         System.out.println("Paid Rs." + amount + " using UPI");
17     }
18 }
19
20 public class PaymentDemo {
21     Run | Debug
22     public static void main(String[] args) {
23
24         Payment p1 = new CreditCardPayment();
25         Payment p2 = new UPIPayment();
26
27         p1.makePayment(amount: 5000);
28         p2.makePayment(amount: 1500);
29     }
```

OUTPUT:

```
[Running] cd "c:\Users\Arpit\java_assignments\" && javac PaymentDemo.java && java PaymentDemo
Paid Rs.5000.0 using Credit Card
Paid Rs.1500.0 using UPI

[Done] exited with code=0 in 2.351 seconds
```

2. Create an abstract FoodOrder class with an abstract method calculateBill(). Implement two subclasses, DineInOrder and TakeAwayOrder, to calculate and display the total bill based on the order type.

```
J FoodOrderDemo.java X
J FoodOrderDemo.java > FoodOrderDemo
1  abstract class FoodOrder {
2
3      abstract void calculateBill();
4  }
5
6  class DineInOrder extends FoodOrder {
7
8      double amount = 500;
9
10     void calculateBill() {
11         System.out.println("Dine-In Bill: Rs." + amount);
12     }
13 }
14
15 class TakeAwayOrder extends FoodOrder {
16
17     double amount = 500;
18     double packingCharge = 20;
19
20     void calculateBill() {
21         System.out.println("Takeaway Bill: Rs." + (amount + packingCharge));
22     }
23 }
24
25 public class FoodOrderDemo {
26     Run | Debug
27     public static void main(String[] args) {
28
29         FoodOrder f1 = new DineInOrder();
30         FoodOrder f2 = new TakeAwayOrder();
31
32         f1.calculateBill();
33         f2.calculateBill();
34     }
35 }
```

OUTPUT:

```
[Running] cd "c:\Users\Arpit\java_assignments\" && javac FoodOrderDemo.java && java FoodOrderDemo
Dine-In Bill: Rs.500.0
Takeaway Bill: Rs.520.0

[Done] exited with code=0 in 2.096 seconds
```